Q-BRIDGE: Verifiable Policy Queries over Joint Confidence Views

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***Abstract*—Recommendation systems often use historical logs and randomized experiments to select policies, rules for assigning items to users, that improve outcomes over a reference. However, observational bias, sampling uncertainty, and limited transfer across populations make policy gains difficult to establish. Evaluating a limited set of candidate policies leaves a decision gap. When none guarantees the desired improvement, it remains unclear whether another policy could meet the target or whether the available evidence rules out such a guarantee for all policies. We study this question by treating policy selection as a family of queries over shared uncertainty in action outcomes. Q-BRIDGE materializes observational intervals and randomized group constraints into a joint confidence view. It supports evaluating individual policies, finding the largest worst-case gain achievable by any policy, and certifying when no policy meets a requested threshold. For disjoint action groups, we develop a three-step algorithm that finds an optimal policy with independently verifiable bounds; exact linear programming handles more general constraints. Our algorithm matches the optimal guarantees computed by exact linear programming across 2,160 synthetic policy optimization problems. In 107 cases where evaluating the original candidate policies leaves the existence of a qualifying policy unresolved, searching the full policy family finds qualifying policies in 86 cases and rules out such guarantees in the remaining 21. Retaining intermediate results reduces the cost of repeated policy queries. Experiments on the Coat recommendation dataset illustrate how the available evidence can support policy redesign or justify stopping the search.**

***Index Terms*—query processing, uncertain data, materialized views, causal inference, verifiable computation**

I. Introduction

A recommendation analyst often uses production logs and randomized experiments to select policies that improve outcomes [1]. A policy specifies how actions are assigned to users. Its gain measures the improvement in expected success over a reference policy.

Estimating this gain is challenging. In production logs, action assignments may depend on unobserved user preferences [2]. Randomized experiments provide more reliable evidence, but population differences can limit transfer [3]. Here, we allow group averages to transfer while individual item outcomes may differ. Sampling uncertainty further limits what can be learned from either source. As a result, the available evidence may support a range of possible action-success probabilities, leading to different gains for the same policy. A policy that appears promising under one set of probabilities may perform poorly under another. This makes it important to identify policies with gains guaranteed across all possibilities consistent with the evidence.

Evaluating a limited set of candidate policies leaves a decision gap. If none achieves the desired guarantee, the analyst faces a further question: could another policy succeed, or is the available evidence insufficient to support the requested guarantee for any policy? Answering this question requires extending evaluation from individual candidates to the entire policy family.

We develop Q-BRIDGE[1], a query framework for evaluating and optimizing policies under shared uncertainty. Q-BRIDGE combines observational intervals and randomized group constraints into a materialized *confidence view*, representing the action-success probabilities consistent with the available evidence. Each policy defines a linear query over this view, with its worst-case gain providing a guaranteed lower bound. Beyond evaluating individual policies, Q-BRIDGE finds the largest worst-case gain achievable by any policy and determines whether the entire family can meet a requested threshold. Independently checkable certificates verify the returned bounds.

Running example. An analyst recommends one of four jackets, A–D, to one user population; success means a high rating. Figure 1 uses their unknown success probabilities $m_A, \ldots, m_D$ and a reference that recommends each with probability $1/4$. Logs bound individual rates; transferred experimental group averages give $(m_A + m_C)/2 = .65$ and $(m_B + m_D)/2 = .70$. The reference's success probability is $(.65 + .70)/2 = .675$. Since $m_A \leq .6$, the first group forces $m_C \geq .7$. Always choosing C guarantees $.7 - .675 = .025 = 2.5$ percentage points (pp). The allowed vector $(.6, .7, .7, .7)$ limits every action to .7, ruling out a larger guarantee. Matching bounds proves optimality.

Figure 1(b) tests a margin $\delta = .001 = 0.1$ pp. Section IV-D works through these three policies to explain when saved evidence suffices and when a new solve is needed.

[1]Code and saved certificate examples: https://github.com/yzd2002/Q-BRIDGE-ICDE2027.

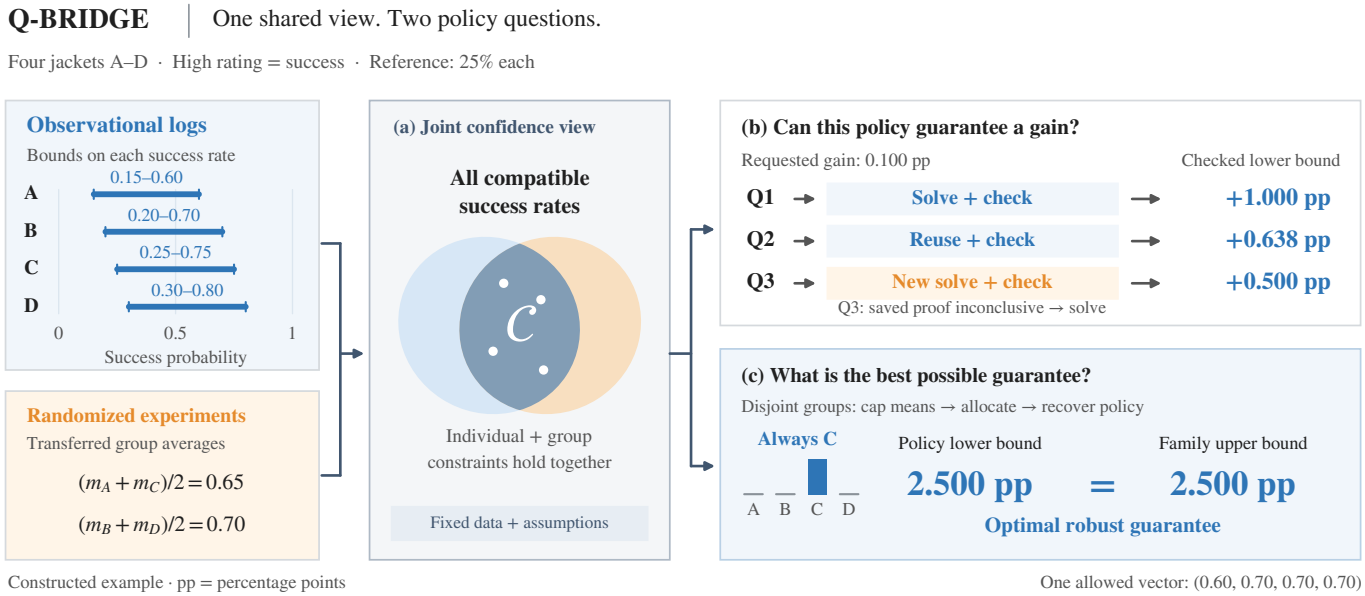


Fig. 1. One view, two query paths. (a) Jointly feasible success rates (schematic intersection). (b) Evaluate a proposed policy: Q2 reuses evidence; Q3 needs a new solve. (c) Optimize over all policies: matching bounds establish the best guarantee, 2.5 pp. This constructed example uses rounded inputs.

One view for changing policies (II–III). Jointly covered constraints support guarantees for all policies, including policies chosen after earlier query answers.

Search beyond the candidates (IV). We prove a three-step construction for disjoint groups and recover certificates against the original constraints. A general exact LP handles overlaps.

Policy benefit and execution cost (V–VI). We separate better policy choice, cheaper repeated queries and stronger source information; Coat connects the answers to recommendation decisions.

Table I summarizes the interface.

II. Policy Queries and Their Guarantees

A policy is useful here when its improvement survives uncertainty about item outcomes. We first define that question, then explain which assumptions turn the two data sources into constraints.

A. The query and the stored object

Let $X \in \{1, \dots, K\}$ identify a user group, $A \in \{1, \dots, L\}$ an action such as recommending a jacket, and $Y \in \{0, 1\}$ success. There are $d = KL$ group–action pairs. The potential outcome $Y(a)$ asks whether this user would succeed if given action a, even when that action was not observed. Observational source O defines the target population; randomized source R may describe another population. Their means are $m^O_{xa} = \mathbb{E}_O[Y(a) \mid X = x]$ and $m^R_{xa} = \mathbb{E}_R[Y(a) \mid X = x]$.

Policy π_{xa} is the probability of taking action a in context x; π_0 is the reference and w_x the target context proportion. The expected gain is

$$\Delta(c) = \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m^O_{xa} = c^\top m^O. \tag{1}$$

Thus $c_{xa} = w_x(\pi_{xa} - \pi_{0,xa})$ records where action probability increases or decreases. Each policy row and w sums to one; both are caller-specified.

A context groups users; a moment group combines actions within one context. Flatten the context–action pairs into $i = 1, \dots, d$. The view stores individual bounds ℓ, u and J group bounds a, b. Each row of T forms a weighted sum of action-success probabilities:

$$\widehat{\mathcal{C}} = \{m : \ell \le m \le u,\ a \le Tm \le b\}. \tag{2}$$

Each m describes one possible set of success probabilities. The view is *joint*: it retains probabilities that satisfy all individual and group bounds together. Figure 1's rows average A/C and B/D, so changing one rate can restrict another.

The *worst-case gain* is $V(c) = \min_{m \in \widehat{\mathcal{C}}} c^\top m$. A query tests $V(c) > \delta$, usually $\delta = .001 = 0.1$ pp. Under the source assumptions, Section III's construction contains all true target rates together with probability at least $1 - \alpha$. Whenever it does, $V(c)$ lower-bounds actual expected gain.

Data and declarations. Events (x, a, y) are independent and identically distributed within each source; repeated observations of one user need user-level treatment (Section VI-D). Counts are N^s_{xay}, with $N^s_{xa} = \sum_y N^s_{xay}$ and $S^s_{xa} = N^s_{xa1}$. Randomization requires $A \perp Y(a) \mid X$ and known $q_{xa} = P_R(A = a \mid X = x) > 0$ on moment supports. A union bound allows dependence across sources and rows.

Prespecify contexts, target, source assumptions, rows, sensitivities and confidence allocation, optionally from independent design data. Objectives and thresholds may adapt; new snapshots or confidence constructions need another guarantee.

TABLE I
THREE POLICY QUESTIONS OVER ONE FIXED CONFIDENCE VIEW.

Question	Answer and independently checked evidence
Does this policy guarantee gain $> \delta$?	True: a lower bound above δ. False: an allowed counterexample. Unknown: neither established. Reuse or solve for this policy; overlapping groups allowed.
What is the largest worst-case gain?	Capacity $\mathcal{R}$ and an attaining policy: its lower bound matches an upper bound for all policies. Our direct construction handles disjoint nonnegative groups; a general exact LP also handles overlaps.
Can every such guarantee be ruled out?	Yes, if one allowed mean vector limits even the best policy's gain to δ. No full optimization is needed when this witness suffices.

B. Bounding observational selection bias

The logged rate $p_{xa} = P_O(Y = 1 \mid x, a)$ describes users who received action a; m^O_{xa} describes what would happen if the target group received it. We bound their difference without estimating hidden preferences U. Let e_O denote assignment probabilities. Assume the recorded outcome equals the outcome under the assigned action, assignment is independent of potential outcomes given (X, U), and these probabilities are positive. Reweighting then gives

$$r^*(x, a, y) = \mathbb{E}_O\left[\frac{e_O(a \mid x)}{e_O(a \mid x, U)}\middle| x, a, y\right],$$
$$m^O_{xa} = \mathbb{E}_O[r^* Y \mid x, a]. \tag{3}$$

The weight r^* corrects over- or under-representation among logged successes and failures. The analyst declares $1/\Lambda \le r^* \le \Lambda$: $\Lambda = 1$ permits no such selection difference; larger Λ widens the bounds. This limits an outcome-conditional average, not assignment odds or every hidden-factor weight. The logs alone do not justify Λ [4].

For binary outcomes, these weight bounds translate directly into a permitted interval for the target mean [5, Eq. (9)]:

$$L_\Lambda(p) = \max\{p/\Lambda, 1 - \Lambda(1 - p)\},$$
$$U_\Lambda(p) = \min\{\Lambda p, 1 - (1 - p)/\Lambda\}. \tag{4}$$

Within one cell, write $p = p_{xa}$, $m = m^O_{xa}$ and $r_y = r^*(x, a, y)$. For $0 < p < 1$, the mean-one condition $pr_1 + (1 - p)r_0 = 1$ gives $r_1 = m/p$ and $r_0 = (1 - m)/(1 - p)$. Bounding both gives (4), exact for this relaxation [6].

Consistency provides additional information. Outcomes under a are known for those who received it; the remaining potential outcomes may be zero or one. Thus

$$P_O(A = a, Y = 1 \mid x) \le m^O_{xa} \le 1 - P_O(A = a, Y = 0 \mid x). \tag{5}$$

These probabilities use all people in context x as the denominator, not just those receiving a. Both Fusion and observational-only (OBS) use these constraints. With exposure .8 and $p = .5$, they give $[.4, .6]$, versus $[.25, .75]$ from $\Lambda = 2$ alone.

C. Which randomized information transfers?

For example, an experiment may support an average rating probability for vests in the target population without supporting the same probability for every individual vest. Row j selects context x_j, actions S_j and nonnegative weights ν_{ja} summing to one. The normalized matrix H has $H_{j,(x_j,a)} = \nu_{ja}$ on this group and zero elsewhere. It is the exact row-normalization of stored T, fixed before inference (III-C). The declaration is

$$\tau_j = (Hm^R)_j, \qquad |(Hm^O)_j - \tau_j| \le \rho_j. \tag{6}$$

Here ρ_j permits source differences in a group mean; zero requires equality. Thus Λ controls selection bias, ρ transfer differences, and α sampling error. Two positive-weight action means can shift by $(h, -\nu_1 h/\nu_2)$ while preserving their average: group agreement does not establish action-level agreement.

If a policy contrast is a weighted combination of the transferred group averages, those averages determine its gain when known exactly: $c = H^\top v$ implies $c^\top m = v^\top \tau$ at $\rho = 0$. Other contrasts may remain uncertain. Assuming every individual item mean transfers ($H = I_d$, $\rho = 0$) is stronger; Coat includes this control.

III. BUILDING ONE CONFIDENCE VIEW

This stage converts the two datasets into reusable bounds, before any policy query is answered. For independent binary events, success/failure counts for each group–action pair suffice. With n_O, n_R source events, aggregation costs $O(n_O + n_R)$ work and $O(d)$ storage; later policies need no rescan. Repeated user observations also require user-level variances (Section VI-D).

A. Observational intervals and an explicit budget

Here a confidence *budget* is permitted failure probability, not sample size. Split $\alpha = \alpha_O + \alpha_R$ between sources. Of α_O, half covers d two-sided Clopper–Pearson (CP) intervals [7] $[p^-_{xa}, p^+_{xa}]$, each with failure probability $\alpha_O/(2d)$. The other half covers $2d$ one-sided joint-event bounds in (5), each at $\alpha_O/(4d)$, with denominator $N^O_x = \sum_{a,y} N^O_{xay}$. Denote their lower endpoints g^1_{xa}, g^0_{xa}. Then

$$\ell_{xa} = \max\{L_\Lambda(p^-_{xa}), g^1_{xa}\},$$
$$u_{xa} = \min\{U_\Lambda(p^+_{xa}), 1 - g^0_{xa}\}. \tag{7}$$

Monotonicity makes propagation conservative. Conditional binomial coverage permits random stratum counts. Empty action cells contribute $[0, 1]$; empty contexts add no joint restriction.

The risk that any bound fails is at most the sum of their failure probabilities, even for dependent bounds.

B. Two prespecified randomized constructions

Both routes below produce an interval $[\tau_j^-, \tau_j^+]$ for the same experimental group mean. Transfer expands it by ρ_j on either side for the target mean; III-C converts this restriction to stored row T_j. Allocate α_R to one route, chosen before outcomes: average individual-mean intervals, or estimate the group mean directly. The synthetic primary comparison uses the former; Coat reports both.

Projected cell intervals. Construct simultaneous binomial intervals $[r_i^-, r_i^+]$ for every randomized cell appearing in H, splitting α_R across those cells. Nonnegativity of H gives normalized group intervals $[Hr^-, Hr^+]$. Only these *group* intervals enter (6); individual experimental means are not equated to target means. Selecting the tighter route after seeing data is not covered by either individual budget.

Group scores. Conditional on an event falling in (x_j, S_j), define $q_j(a) = q_{x_j a}/\sum_{b\in S_j} q_{x_j b}$ and $Z_j = \nu_{jA} Y/q_j(A)$. This reweights experimental assignment to the desired group weights: $\mathbb{E}[Z_j] = \tau_j$ and $0 \le Z_j \le B_j = \max_a \nu_{ja}/q_j(a)$. Let n_j count these events and $\bar{Z}_j$ be their average score. Counts, sums and squared sums follow from N^R_{xay}, since $Y^2 = Y$. If every multiplier $\nu_{ja}/q_j(a)$ equals one, $Z_j = Y$ and we use a binomial interval. Otherwise, for $n_j \ge 2$, the empirical Bernstein interval [8] has half-width

$$h_j = \sqrt{\frac{2\widehat{\mathrm{Var}}(Z_j)\log(4/\beta_j)}{n_j}} + \frac{7B_j\log(4/\beta_j)}{3(n_j-1)}. \tag{8}$$

Here $\beta_j = \alpha_R/J$, and the sample variance uses $n_j - 1$. The two terms account for score variation and its bounded range. The interval is $[\bar{Z}_j - h_j, \bar{Z}_j + h_j] \cap [0,1]$: only the expectation interval is clipped, not the scores. Insufficient observations use the full range. The union bound covers overlapping groups sharing events.

C. Keeping stored constraints faithful to the bounds

Rounding must not exclude a probability vector that the statistical bounds allow. The checker therefore reads stored binary64 numbers as exact fractions. For example, five stored values of .2 sum to $1 + 2^{-54}$, not exactly one; we account for that difference when storing a group average.

For nonzero T_j, its exact mass $\kappa_j = \sum_i T_{ji}$ defines $H_{ji} = T_{ji}/\kappa_j$ and $T_j m = \kappa_j H_j m$; zero rows impose deterministic zero constraints. Assignment probabilities are either exact or explicitly normalized nonnegative design weights.

Scale by κ_j and round outward so storage cannot narrow the permitted interval:

$$\begin{aligned} a_j &= \mathrm{down}\left[\kappa_j(\tau_j^- - \rho_j)\right], \\ b_j &= \mathrm{up}\left[\kappa_j(\tau_j^+ + \rho_j)\right]. \end{aligned} \tag{9}$$

Projected intervals use exact $T_j r^-$ and $T_j r^+$ expanded by $\kappa_j\rho_j$. Observational endpoints, complements, score moments and normalization use rational arithmetic with one outward endpoint rounding; exact empty intervals are refused first. The statistical layer trusts CP quantiles and floating-point Bernstein arithmetic, including logarithms and square roots.

D. Why one view supports many queries

Set $\alpha_O + \alpha_R = \alpha$ (equal halves for fusion here). Once the true vector lies in the view, changing the policy changes only the objective; no new coverage event or confidence budget is needed.

Theorem 1 (One event, all objectives): Under the source and sampling conditions in Section II, $P(m^O \in \widehat{\mathcal{C}}) \ge 1 - \alpha$. On that event, every linear objective, including one chosen adaptively from previous answers, satisfies $V(c) \le c^\top m^O$. Any verified lower approximation to $V(c)$ shares this inequality. A method that cannot establish a usable view or bound makes no positive assertion.

Proof. Conditional on independent design information, interval coverage averaged over stratum counts and monotone mapping cover the box at $1 - \alpha_O$. Prespecified randomized intervals cover Hm^R at $1 - \alpha_R$; transport and outward mapping give target moments. A union bound covers m^O at $1 - \alpha$. On that event, minimizing over the view bounds every objective simultaneously, including adaptive choices. Weak duality extends this to checked lower bounds. □

Reuse retains the statistical design [9], [10]. View identity binds count digests, confidence allocation, assignment design, radius, tolerances and matrix; protocols retain provenance.

IV. Query Execution and Verification

The query engine returns a numerical answer together with a short, independently checkable proof, called a *certificate*. The optimizer produces it; a separate checker verifies it against the stored view $\mathcal{C} = \widehat{\mathcal{C}}$. We evaluate one policy (A), search all policies (B–C), then reuse proofs to answer new queries faster (D–E).

A. Checking a proposed policy

First, a solver proposes an allowed probability vector and the checker verifies it as a *feasible point*. Otherwise the view is marked `uncertain` and no decision is made. This prevents conclusions from contradictory constraints; failure to find a point is not proof that none exists, particularly for sets with no interior.

Together with a lower certificate, a checked feasible point brackets the optimum [11, Sec. 5.5.1]:

$$L(c) \le V(c) \le U(c) = c^\top \widetilde{m}. \tag{10}$$

If $L > \delta$, the policy is guaranteed to pass. If $U \le \delta$, the feasible point is a counterexample to that guarantee. If the bounds straddle the threshold, the answer is unknown. This tests a guarantee, rather than the actual effect. Saved points remain feasible for later policies, so their new payoffs can be checked without solving again.

How a lower certificate works. Give the lower and upper group constraints nonnegative weights η_L, η_U. For any feasible m, both $\eta_L^\top(a - Tm)$ and $\eta_U^\top(Tm - b)$ are nonpositive. Adding

them to the gain can only lower it. With $z = c + T^\top(\eta_U - \eta_L)$, minimizing this adjusted expression over the individual intervals gives

$$D(c;\eta) = a^\top \eta_L - b^\top \eta_U + \sum_i \min\{z_i \ell_i, z_i u_i\} \le V(c). \quad (11)$$

Each term chooses the lower-cost endpoint: ℓ_i when $z_i \ge 0$, u_i otherwise. These constraint weights are the *dual certificate*; a feasible probability vector is a *primal witness*. Checking either needs simple arithmetic, not an optimization replay. Any nonnegative weights give a valid lower bound; better weights tighten it [12].

The checker uses exact arithmetic and rounds lower bounds downward, binding the objective, threshold and view digest. Failed or nonfinite proposals cannot authorize success. Independent optimization certificates are established [13].

B. Certifying a whole policy family

Testing one policy can miss a better one. The family Π contains all randomized policies on the available actions; its *capacity* is the largest worst-case gain. Fix a nonempty compact convex view $\mathcal{C} \subseteq [0,1]^{KL}$, exact probability weights $w \ge 0$, $\sum_x w_x = 1$, and reference $\pi_0 \in \Pi$. Stars will denote returned optimizers. Define

$$\mathcal{R}(\mathcal{C}) = \max_{\pi \in \Pi} \min_{m \in \mathcal{C}} \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m_{xa}, \quad (12)$$

$$F(m) = \sum_x w_x \left[\max_a m_{xa} - \sum_a \pi_{0,xa} m_{xa} \right]. \quad (13)$$

The inner minimum gives one policy's worst case; the outer maximum chooses the best policy. The reference ensures $\mathcal{R} \ge 0$. At one allowed m, $F(m)$ chooses the best action per context. It bounds $\mathcal{R}$ because each policy's minimum cannot exceed its payoff at this point. Thus $U(c)$ bounds one policy's guarantee; $F(m)$ bounds every policy's guarantee. A checked $F(m) \le \delta$ excludes all above-threshold guarantees, not actual effects.

Compact convex domains and a continuous bilinear payoff permit minimax exchange [14], giving $\mathcal{R}(\mathcal{C}) = \min_{m \in \mathcal{C}} F(m)$. We therefore seek an allowed mean vector giving the best action as little advantage as possible, and a policy whose lower certificate reaches this upper bound.

C. Exact capacity for partitioned moments

The direct algorithm applies when each action belongs to at most one nonzero group constraint; actions may also be ungrouped. Groups cannot cross user contexts, and their weights must be nonnegative. We solve each context separately. Minimize $t - p^\top m$: the largest action mean minus reference payoff. Here t caps every m_i and $p = \pi_{0,x}$ is the reference distribution. Steps 1–2 find allowed rates that limit every policy's guarantee; Step 3 finds a policy whose gain is at least this limit in every allowed scenario. Minimax and allocation are classical [15], [16]; our construction proves that the smallest feasible cap suffices and recovers an original-constraint certificate.

Theorem 2 (An attaining policy with an original-view certificate): Let $\mathcal{C}$ be a nonempty rational box-and-moment view with $0 \le \ell \le u \le 1$. Suppose every nonzero moment row is nonnegative, belongs to one context, and has positive support disjoint from every other row. For any rational probability weights w and reference policy π_0, including zero entries, the three-step construction returns $m^* \in \mathcal{C}$, $\pi^* \in \Pi$ and nonnegative original-row multipliers η_L, η_U such that, with $c^*_{xa} = w_x(\pi^*_{xa} - \pi_{0,xa})$,

$$D(c^*;\eta) = c^{*\top} m^* = F(m^*) = \mathcal{R}(\mathcal{C}).$$

The dense implementation uses $O(dJ + d^2)$ exact arithmetic/comparison operations; the independent receiver uses $O(dJ + d)$. These are unit-cost rational-operation counts, excluding coefficient-dependent bit costs, encoding and hashing.

A policy lower bound cannot exceed capacity, and $F(m^*)$ cannot fall below it. Matching them proves optimality independently of the optimizer. Unsupported rows are rejected; zero rows require $a_j \le 0 \le b_j$ and receive zero multipliers. Ties use index order.

Step 1: Find the smallest feasible cap.

Here $p = \pi_{0,x}$ denotes reference probabilities, not logged success rates. The cap must leave enough total success probability to meet each group's lower bound. Define

$$t_\ell = \max_i \ell_i, \qquad g_j(t) = \sum_i T_{ji} \min(u_i, t),$$
$$t_0 = \min\{t \ge t_\ell : g_j(t) \ge a_j \quad \forall j\},$$
$$h_i = \min(u_i, t_0).$$

Each g_j is piecewise linear between upper-endpoint breakpoints; its required root, together with t_ℓ, determines t_0. Upper group bounds cannot force a higher cap. Because the view is nonempty and groups are disjoint, each can choose mass in $[\max(a_j, T_j\ell), \min(b_j, T_j h)]$. Hence the capped view is feasible and its maximum is t_0. In Fig. 1, $t_0 = .7$ and $h = (.6, .7, .7, .7)$.

Minimum-cap sufficiency. A larger cap could raise the reference payoff, so feasibility alone is insufficient. For feasible m with $t = \max_i m_i$, clip to $v_i = \min(m_i, t_0)$, preserving lower boxes and upper moments. Restore violated lower moments within $[v, h]$ to $a_j \le b_j$; disjoint supports prevent interference, giving feasible $v' \ge v$. Clipping lowers the maximum by $t - t_0$ and the reference average by at most that amount: $p^\top(m - v) \le t - t_0$. Repairs only raise the average. Hence

$$\max_i v'_i - p^\top v' \le t_0 - p^\top v \le t - p^\top m.$$

Thus maximizing $p^\top m$ within this cap minimizes the original gap.

Step 2: Allocate means within the cap. With the maximum fixed at t_0, increasing the reference payoff reduces the gap. We allocate *means*, not policy probabilities. Raising m_i by ϵ costs group mass $T_{ji}\epsilon$ and gains $p_i\epsilon$. Starting at ℓ, fill each nonzero row in decreasing p_i/T_{ji} order on $T_{ji} > 0$, to total mass $\min(b_j, T_j h)$; uncovered coordinates take h_i. The added mass is $\min(b_j, T_j h) - T_j \ell$. Step 1 ensures the final total

$\min(b_j, T_j h)$ meets a_j. Nonnegative weights justify filling even zero-ratio coordinates.

To build the certificate, record an upper-constraint multiplier λ_j per unit of group mass. Use zero when $b_j \geq T_j h$, including equality; the cap already limits the total. Otherwise use the largest support ratio if $b_j = T_j \ell$, or the last ratio receiving positive mass, including exact boundary exhaustion. After subtracting these multiplier terms, $\xi = p - T^\top \lambda$ measures the remaining reference-payoff gain from raising each mean. Its sign explains the endpoint choice:

$$\xi_i > 0 \Rightarrow m_i^* = h_i, \quad \xi_i < 0 \Rightarrow m_i^* = \ell_i,$$
$$\lambda_j(b_j - T_j m^*) = 0.$$

A positive multiplier requires a tight upper bound. The example gives $m^* = (.6, .7, .7, .7)$, $\lambda = (0, 0)$ and family upper bound .025.

Step 3: Recover the policy and certificate. Choosing a best action at m^* alone is unsafe: B ties for best there, but allowed rates $(.6, .6, .7, .8)$ give pure B gain $-.075$. We must replace the artificial cap by a policy and original-constraint multipliers. Write the lower and upper multipliers as $\eta_L = \eta$ and $\eta_U = \lambda$. For equality in (11), the policy must select only maxima of m^*, rows with positive multipliers must be tight, and $z = \pi - p + T^\top(\lambda - \eta)$ must satisfy $z_i > 0 \Rightarrow m_i^* = \ell_i$ and $z_i < 0 \Rightarrow m_i^* = u_i$.

Start by canceling positive residuals at actions that can reach the cap; any missing probability will be supplied by a constraint that forces the cap:

$$\widetilde{\pi}_i = (\xi_i)_+ \mathbf{1}\{u_i \geq t_0\}, \qquad D_\pi = 1 - \sum_i \widetilde{\pi}_i \geq 0.$$

Here $(\xi_i)_+ = \max(\xi_i, 0)$. Since $0 \leq \widetilde{\pi}_i \leq p_i$, the deficit D_π is nonnegative. Initial coefficients $\widetilde{\pi}_i - \xi_i$ already have the required endpoint signs. In the example, initialization gives $(0, .25, .25, .25)$: the remaining .25 will go to C, the only action in the cap-forcing A/C group that can reach .7.

If $t_0 = t_\ell$, add D_π to a coordinate with $\ell_i = t_0$ and set all lower multipliers η to zero; its box coefficient increases at this lower endpoint. Otherwise choose the first row whose required root is t_0, so it forces the cap. Since $T_j h = a_j$, its supported means must equal h_i, and the zero-multiplier choice in Step 2 gives $\lambda_j = 0$, even if its upper moment is also tight. Set

$$E = \{i : T_{ji} > 0,\ u_i \geq t_0\}, \qquad \sigma_E = \sum_{i \in E} T_{ji} > 0,$$

where positivity follows because otherwise this row would attain its bound below t_0. Set $\eta_j = D_\pi / \sigma_E$, with other lower multipliers zero. Add $\eta_j T_{ji}$ to $\widetilde{\pi}_i$ on E. Updated box coefficients are zero on E and $-p_i - \eta_j T_{ji} \leq 0$ elsewhere in this row, where $m_i^* = h_i = u_i$; other coefficients are unchanged. The non-strict $u_i \geq t_0$ includes roots at breakpoints.

The completed policy sums to one and selects only maxima. The endpoint signs and tight rows with positive multipliers make $D(\pi - p; (\eta, \lambda)) = (\pi - p)^\top m^* = t_0 - p^\top m^*$. Multiplying each context's objective and both multiplier types by w_x and concatenating gives the theorem. Zero-weight contexts still need feasible means and legal policies. The checker validates original constraints, probabilities, weight signs and matching bounds; it never trusts the artificial cap. □

For the example's remaining mass, $\sigma_E = .5$, $\eta_L = (.5, 0)$ and $\eta_U = 0$, giving $\pi^* = (0, .25, .5, .25)$. The lower bound $.65(.5) - .6(.5) = .025$ equals the family upper bound $.7 - .675 = .025$. Pure C in Fig. 1 is another optimal policy. These decimal teaching inputs differ from represented inputs checked in measured runs.

Stopping without the full construction. An allowed vector with equal action-success probabilities within each context makes every policy tie with the reference. This offers a cheap stopping certificate, including for overlapping groups.

Proposition 1 (Zero capacity): If all $w_x > 0$ and $\pi_{0,xa} > 0$, then $\mathcal{R}(\mathcal{C}) = 0$ exactly when $\mathcal{C}$ contains a vector constant across actions within each context. Without strict positivity, a constant point is sufficient but need not be necessary.

Proof. The nonnegative summands of F vanish, under strict positivity, exactly at constant rows. Compactness attains its minimum; the minimax identity gives the result. □

For nonnegative single-context moment rows, even overlapping ones, form

$$I_x = [\max_a \ell_{xa}, \min_a u_{xa}] \cap \bigcap_{j : x_j = x} [a_j/\kappa_j, b_j/\kappa_j].$$

Here $\kappa_j = \sum_i T_{ji} > 0$; zero rows require $a_j \leq 0 \leq b_j$. Choose $v_x \in I_x$, set every $m_{xa} = v_x$, and check the original constraints. Under strict positivity, an empty intersection proves $\mathcal{R} > 0$, but not $\mathcal{R} > \delta$ for positive δ.

Saved policy lower bounds (including reference zero) and values $F(m)$ at saved feasible points also bracket capacity. They certify existence or exclusion if the corresponding bound crosses δ; otherwise search continues. Reuse requires identical $\mathcal{C}, w, \pi_0$. Section V-C separates loose bounds from restricted policy choice.

D. Reusing certificates across policies

A new policy changes c, but leaves the view constraints unchanged. Old nonnegative constraint weights therefore still give valid lower bounds, though not necessarily tight ones. The bank normalizes saved weights by their old objective's binary64 ℓ_1 norm; the new norm supplies scale s. We retain their group contribution $k = a^\top \eta_L - b^\top \eta_U$ and coefficient adjustment $r = T^\top(\eta_U - \eta_L)$. For stored $s \geq 0$,

$$D(c; s\eta) = sk + \sum_i \min\{(c_i + sr_i)\ell_i, (c_i + sr_i)u_i\}. \quad (14)$$

Thus later queries reuse the matrix–weight product r instead of recomputing it. Setup costs $O(\mathrm{nnz}(T) + d + J)$ exact operations, later checks $O(d)$; bit costs depend on coefficients. A fresh checker reconstructs the saved products exactly, using stored base weights and scale rather than separately rounded multipliers. Dual and bounded-Lagrangian reuse are established [17], [18], [19]; persistent LP reoptimization supplies a control [20].

The example: solve once, then try reuse. Q1 changes the uniform policy to $(.15, .15, .35, .35)$, giving $c = (-.1, -.1, .1, .1)$. The two group means imply

$$\Delta_1 = .27 - .2(m_A + m_B) \geq .01.$$

The allowed vector $(.6, .7, .7, .7)$ attains it; interval-only and group-only bounds are $-.075$ and $-.13$. Q1's weights $\eta_L = (.2, .2), \eta_U = 0$ give $k = .27$, $r = (-.1, -.1, -.1, -.1)$.

Q2 has $c = (-.105, -.09, .105, .09)$ and norm .39, versus Q1's .4. Scaling Q1's unnormalized weights by $.39/.4 = .975$ is equivalent to the bank's normalization. Equation (14) gives .006375, passing .001 without finding the optimum .0105. Q3 has $c = (-.05, -.15, .15, .05)$ and scale 1. Reuse gives $-.0225$, inconclusive; a new solve gives .005. A weak lower bound triggers work, not rejection.

E. Execution choices and numerical safeguards

After validating the view, the engine tries saved bounds before solving a new linear program (LP). It checks the zero-weight bound, the last solver weights, and feasible points, with a first-in, first-out bank of at most 16 candidates of each enabled kind. A cache handles an identical objective; the bank handles a different policy. Zero-norm old objectives are not stored; a zero new norm uses scale zero. Solver cutoffs require exact checking, and undecided requests resume without a cutoff [21].

Two optional features reduce arithmetic. *Retention* saves the exact terms k, r in (14). *Screening* first bounds a candidate's answer using cheap floating-point intervals and defers it if it cannot decide the query. Screening never authorizes an answer; overflow triggers exact work. Before a false answer or a solver call, deferred candidates are replayed as needed to preserve the unscreened plan's evidence and solver path. Reuse hits retain the last solver state; zero-weight answers are not counted as reuse. The artifact specifies the replay order and cutoff handling.

Structural controls use box endpoints, continuous knapsack for single rows [22], or sparse HiGHS for overlaps [23]; only the reconstructed global dual is certified.

From stored contrasts to intended policies. If intended rational c^{ideal} is stored as c, let $e = c^{\text{ideal}} - c$. Because $0 \leq m_i \leq 1$, the objective change lies between $\sum_i \min(e_i, 0)$ and $\sum_i \max(e_i, 0)$. Applying these corrections to the checked *minimum*-gain bracket gives

$$[L + \sum_i \min(e_i, 0),\ U + \sum_i \max(e_i, 0)]. \tag{15}$$

The additions and threshold test are exact; a positive stored bound need not certify an ideal zero-margin query. Fixed-query timings (V-D) use stored contrasts, family queries (V-B) exact probabilities, and Coat (VI-D) both with (15) as needed.

V. Evaluation Design

We test three questions: can family search find guarantees beyond the initial candidates (VI-A); can reuse reduce cost with the same evidence (VI-B); and can fusion strengthen a fixed policy's guarantee (VI-C)? Synthetic outcomes are known; Coat provides real user ratings (VI-D). Family analysis is retrospective; the source-information endpoint was registered before analysis. Exact-solver and new Coat timings use an AMD EPYC 7763, Python 3.12.3 and one CPU/native thread.

A. Fresh worlds and policy design

A simulated *world* fixes user groups, assignment and success probabilities. Five fresh datasets per world separate sample variation from population differences. The registered study draws 48 confounded worlds: $(K, L) \in \{(2, 4), (8, 8), (16, 16)\}$, uniform/skewed randomization and group drift $0/.02$, with four worlds per stratum and five inference samples per world. Twelve additional worlds test near-unit sensitivity. A latent binary factor affects exposure and rewards. Metadata $h_a \in [-.5, .5]$ have a designed positive reward association and are fixed before inference. Random groups permit individual shifts while controlling group drift.

Inference uses $2{,}000d$ O events and $100d$ or $1{,}000d$ R events. Separate design data ($500d$ O and $100d$ R events) choose three candidates before the inference data are seen. *Metadata* favors actions with higher fixed scores, using probabilities $\mathrm{softmax}(1.5h)$. *R heuristic* favors larger design-experiment success rates, using $\mathrm{softmax}(4\widehat{p}_x)$ with $\widehat{p}_{xa} = (S_{xa} + .5)/(N_{xa} + 1)$. Softmax normalizes exponentiated scores to sum to one. *Robust design* selects the largest checked positive lower bound among twelve policy tilts, or keeps the uniform reference if none qualifies. Tilts use linear or alternating scores with slopes $\{-2, -1, -.5, .5, 1, 2\}$; ties retain the first maximum. Design outcomes never enter inference counts. We compare $\Lambda = 2$ with an oracle boundary computed from simulator truth and rounded upward. Counts and policy rules stay fixed; the design-selected policy can vary with radius.

These three candidates are not the unrestricted family search. We compare three sources of constraints: *OBS* uses only logged data; *R* uses only transferred experimental group information; *Fusion* combines both. *Projected* averages action-level intervals; *group* estimates group averages directly (III-B). Each standalone method uses the full failure budget α; Fusion splits it between sources. The primary endpoint compares projected Fusion with certification by either full-α OBS or projected R, for the metadata policy, low R budget and $\Lambda = 2$. Average paired certification differences within each of 48 independent worlds. Their values lie in $[-1, 1]$, so the one-sided 95% Hoeffding lower bound is the mean minus $\sqrt{2\log(20)/48}$. The registered target requires this lower bound to exceed 0.05, i.e., a five-percentage-point increase in certification frequency. Other settings and the stratified t interval are descriptive. The standalone union counts a case when either source certifies it; it is not a jointly calibrated decision rule.

B. Whole-family queries on fixed views

We reuse the 48 main worlds with five samples each and target $\delta = .001$. Three source constructions (OBS,

projected R, projected Fusion), two R sample sizes, two radii and finite/population information give 24 comparison cells. Population views remove sampling noise using the known simulator statistics, while retaining uncertainty allowed by the source assumptions. The 5,760 comparison positions reduce to 2,160 distinct exact $(\mathcal{C}, w, \pi_0)$ problems, each solved by both methods. Deduplication changes computation, not the compared positions.

The primary recovery study searches 107 of 240 finite-Fusion views unresolved by saved evidence, keeping data and thresholds fixed. We first normalize and recheck the three saved policies exactly. This retrospective search uses Theorem 1's coverage for adaptive policy selection, rather than a new out-of-sample test.

C. A complete exact solver and the source of recovery

SoPlex 8.1.0 in rational mode [24], [25], [26] solves the epigraph LP on identical $(\mathcal{C}, w, \pi_0)$:

$$\min_{m\in\mathcal{C},t} \sum_x w_x\big(t_x - \pi_{0,x}^\top m_x\big), \qquad m_{xa} \le t_x, \quad t_x \text{ free.} \tag{16}$$

Stationarity of free t_x gives $\sum_a \mu_{xa} = w_x$ for its nonnegative epigraph multipliers. Thus $\pi_{xa} = \mu_{xa}/w_x$ is a policy when $w_x > 0$; use the reference row when $w_x = 0$. Original-row duals and a feasible point complete the partition method's certificate contract. Import/export and checking are exact; neither the construction's answer nor a saved point is supplied.

A value-independent hash rule selects 78 distinct requests for three paired timing rounds, with alternating method order. Each method starts with a fresh model and worker. Service time covers transmission after worker initialization through certificate checking, including serialization; startup/cleanup are reported separately. Limits are one CPU thread, ten seconds per service and 8 GiB address space. Failures retain their costs and denominators. Repeated rounds measure machine variation, not new datasets.

To determine whether recovery comes from better bounds or a different policy, we solve the original three policies exactly, obtaining optima L_i^*. Set $L_3^+ = \max(0, L_1^*, L_2^*, L_3^*)$, with zero from the reference. Here L_{saved} is the maximum of zero and saved lower bounds for these same three policies. Then

$$\mathcal{R} - L_{\text{saved}} = \underbrace{L_3^+ - L_{\text{saved}}}_{\text{certificate-transfer gap}} + \underbrace{\mathcal{R} - L_3^+}_{\text{three-policy restriction}} . \tag{17}$$

Both terms are nonnegative. Saved duals solve their stored objectives, but exact policy normalization can loosen transferred bounds. Missing optima leave the decomposition unknown. The 107-case recovery set and threshold remain unchanged.

Coat comparisons. Item gender, jacket type and color define separate partitions. We also test their combined, overlapping constraints. Subviews of one parent retain its bounds and inherit its joint coverage event. Across distinct parent constructions or assumption declarations, no simultaneous 95% guarantee is asserted.

D. Repeated fixed-policy queries

The 63 inputs comprise three primary 1,000-query streams on a 512-cell view; 54 geometry streams varying size ($d = 64, 512, 2048$) and row overlap; and six generated-log streams with 64 actions, one million O and four million R events. The latter two groups contain 100 queries per stream. Queries vary nearby and unrelated policy directions at $\delta = .001$. The primary streams have one solve and 999 reuse hits, all true; other streams include false and unknown answers. These are controlled workloads, not estimates of online hit rates.

Four plans turn screening S and retention P independently on or off. They use identical solvers, candidate order, certificates and checker; their answers, evidence and solver paths must match. The optimized implementation retains decoded views, feasible points and within-query number conversions, and skips zero coefficients. Thus the retention comparison isolates saving the exact terms in (14). A separate "last-first" control changes candidate order, so its certificates may differ.

Joint service includes initialization, validation, queries, failures, solver fallback, serialization and checking with a 16-entry bank. It excludes ingestion, disk I/O, compression, network transfer and separate archival audits. A quality pass checks 45,000 responses; five shuffled timing rounds contribute 225,000 more. Each fresh run uses one CPU/native thread. Ratios compare enabled/disabled cost on the same input with the other switch fixed; group summaries give each input equal weight in a geometric mean. Rounds reflect machine variation. Individual costs, slow cases and all four conditional comparisons are archived.

VI. Experimental Results

A. What complete family optimization adds

Broader policy search changes the fixed-view answer. The decision sequence is simple: of 240 primary views, earlier checks already find a qualifying policy in 12 and rule out every qualifying policy in 121 (zero capacity). Full-family search resolves the remaining 107: it finds a different qualifying policy in 86 and rules out the target in 21, whose best guarantee is positive but at most δ. Thus 98 views support the target and 142 do not. Exact reoptimization of the original three policies recovers none of the 86: their best value, including the reference, is zero in every case. Recovered guarantees have median 0.991 pp, interquartile range $[0.583, 1.549]$ pp and range $[0.105, 3.632]$ pp.

The construction and exact LP agree on all 2,160 distinct synthetic problems. Across family, fixed-policy, Coat and timing checks, all 12,770 responses pass, without timeout or incomplete call.

Exact answers with dimension-dependent cost. Across three rounds, construction/SoPlex geometric-mean service ratios are 0.808–0.835 at $d = 8$, 1.066–1.129 at $d = 64$, and 1.170–1.196 at $d = 256$. The construction is slower in 5/78, 55/78 and 54/78 pairs, respectively. These compare checked Python rational construction with compiled rational LP. Table II gives absolute costs. At $d = 256$, verification alone takes

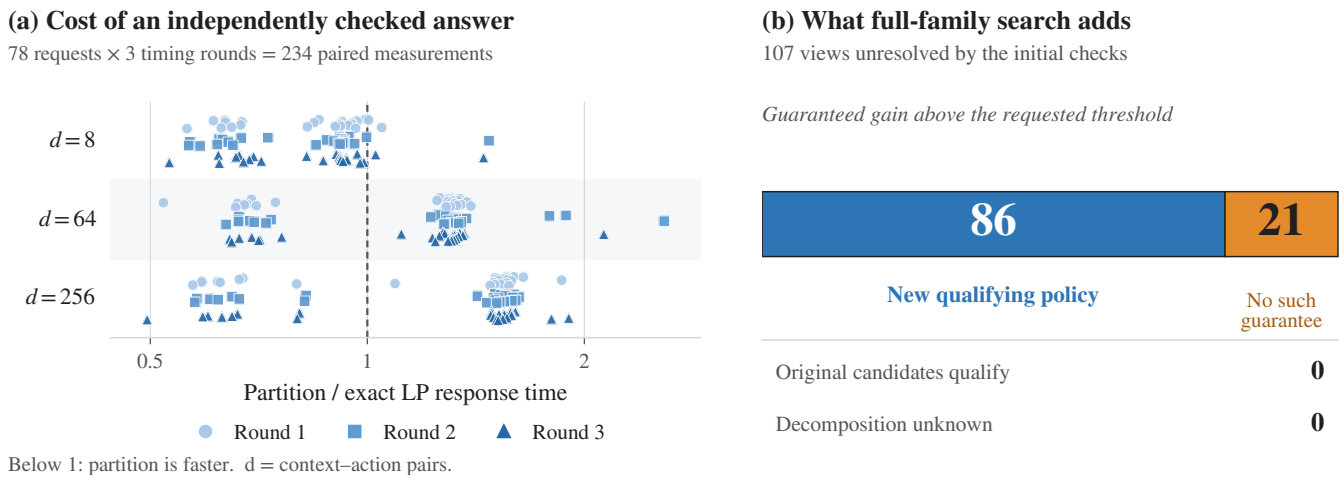


Fig. 2. Checked cost and full-family search. (a) 26 requests per dimension × three dimensions × three rounds: all 234 pairs verify. Response time includes proof checking; offsets separate points. (b) Full-family search resolves all 107 remaining views (of 240): 86 admit a qualifying policy; 21 cannot support the guarantee. Exact rechecking of the original candidates recovers none. The decomposition separates tighter bounds from choosing a different policy. Rounds measure machine variation, not confidence intervals.

TABLE II
ABSOLUTE COST AND PROOF SIZE OF A CHECKED ANSWER.

d	Method	Response (ms)	Check (%)	Whole (ms)	Proof (kB)
8	Part	4.66	24.4	122.2	3.47
8	LP	4.99	22.1	135.2	3.47
64	Part	53.58	23.2	172.2	20.93
64	LP	40.85	31.6	170.0	20.91
256	Part	648.13	22.8	770.3	149.51
256	LP	419.97	34.6	554.0	149.44

Separate medians over 26 requests × 3 rounds per method/dimension. d: context–action pairs; Part: partition construction; LP: SoPlex. Response: worker readiness through checking; Check: verification time as a percentage of response time; Whole: includes startup/cleanup; kB: 1,000 bytes. Paired ratios: Fig. 2(a).

median 144.3/144.6 ms (Part/LP); certificates occupy about 149.5 kB.

Source information is a separate comparison. At the larger R budget with conservative radius, projected R passes 223/240 thresholds versus Fusion's 218/240; R has greater capacity in 128 pairs, Fusion in 100, with 12 ties. At the smaller R budget with oracle-boundary radius, OBS and Fusion both pass 239 thresholds, but OBS has greater capacity in 219 pairs and Fusion in 21. Fusion protects both sources within the same total error budget, so its component intervals can be wider; VI-C separates this cost from the gain due to added constraints. Generating means satisfy all 2,880 finite partition views, and every returned policy assertion agrees with stored truth.

B. The cost of answering repeated policy queries

Retention saves the exact group terms that a new query would otherwise recompute; screening first tests whether an exact check can be deferred. All 270,000 responses pass independent exact replay; core plans match evidence, decisions and native paths. Across the five primary rounds (Fig. 3), retention ratios are 0.735–0.748 without screening and 0.766–0.791 with it. Screening ratios are 1.103–1.138 without retention and 1.165–1.186 with it. Screening also costs more in every geometry and source-derived group-round mean. Its earlier benefit without retention disappears under optimized arithmetic; two of three registered directions hold in every primary round.

The last-first bridge costs 0.833–0.882 times S_0P_1 on primary streams, but is slower in 1/15 primary, 66/270 geometry and 17/30 source-derived input-round pairs. Retention also has adverse individual cases despite lower group means. Thus retention lowers group-average cost without guaranteeing an individual speedup. Candidate order changes only in the bridge.

a) Auxiliary execution controls: Additional tests check stronger efficiency claims. Screening plus retention does not reach the registered cost-ratio target ≤ 0.8 against each feature alone; screening slows every broader-group round. A larger bank also misses its speed target and changes decisions. Materialization speeds ingestion, but preparation costs 1.580/1.619 times hot execution under uniform/skewed assignment. Full controls are archived.

C. Source information and query guarantees

Table III follows three policies chosen before inference (48 worlds ×5 samples each); VI-A instead searches for a better policy on each fixed view. All 60 worlds and 1,200 inference configurations complete. For the metadata policy at $100d$ randomized events and $\Lambda = 2$, fusion certifies 12/240 versus six for the standalone union (OBS six, R zero): six additions, no losses, certification-rate difference 0.025 (2.5 pp). The one-sided Hoeffding lower bound, -0.3283, does not meet the required 0.05 increase; the descriptive t interval is $[-0.0249, 0.0749]$. These are changes in certification fre-

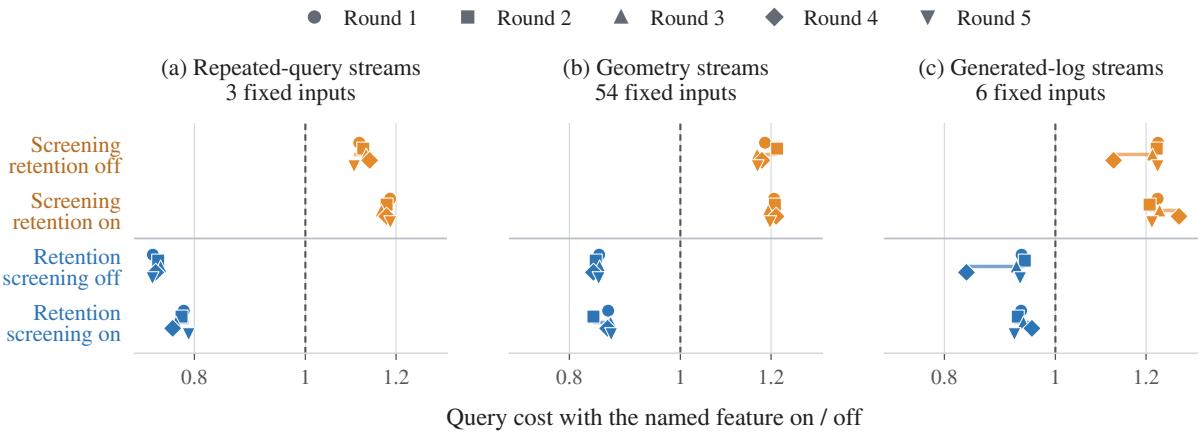


Fig. 3. Retention lowers group-average cost; screening does not. Fixed inputs are (a) three 1,000-query primary streams, (b) 54 geometry streams and (c) six generated-log streams, each of the latter two containing 100 queries. Rows toggle one feature, fixing the other. Points: equal-input geometric means of on/off cost ratios. Lines: five-round ranges, not confidence intervals. Below one is faster.

TABLE III
FIXED-POLICY OUTCOMES UNDER PROJECTED FUSION.

Policy	Certified	Sampling-limited	Ambiguity-limited	True gain $\leq \delta$
Metadata	12	33	195	0
R heuristic	4	43	190	3
Robust design	2	0	0	238

240 queries per policy with $100d$ randomized events, $\Lambda = 2$, and $\delta = .001$ (0.1 pp). Sampling-limited: exact population statistics would suffice. Ambiguity-limited: even those statistics do not suffice despite true gain $> \delta$. "Robust design" selects one candidate before inference; it is not full-family optimization. No case is missed solely by optimization.

quency, not policy gains. All six additions occur in two zero-drift 2×4 worlds.

Larger-budget projected fusion certifies 15 metadata queries, 16 heuristic queries and two robust-design selections. More randomized traffic does not resolve transfer or radius uncertainty.

Why adding a source need not improve the bound. For the same policy contrast c, fusion adds group constraints but splits the confidence budget. Let $\mathcal{B}_\alpha \subseteq \mathcal{B}_{\alpha/2}$ be full-budget OBS and fusion's observational box, with objective minima $B_\gamma = \min_{m \in \mathcal{B}_\gamma} c^\top m$. For nonempty $\mathcal{C}_F = \mathcal{B}_{\alpha/2} \cap \{a \leq Tm \leq b\}$, its optimum obeys

$$V_F - B_\alpha = \underbrace{(V_F - B_{\alpha/2})}_{G \geq 0} - \underbrace{(B_\alpha - B_{\alpha/2})}_{P \geq 0}. \tag{18}$$

Fusion improves the full-budget observational bound exactly when its constraint gain G exceeds its budget cost P.

Proposition 2 (A checkable zero-gain condition): Let $F_c = \arg\min_{m \in \mathcal{B}_{\alpha/2}} c^\top m$. Then $G = 0$ if and only if F_c contains a point satisfying all randomized group constraints. If F_c is a singleton, checking its moment rows decides whether $G = 0$ or $G > 0$.

Proof. A feasible box minimizer attains the box bound in $\mathcal{C}_F$. Conversely, compactness gives an optimizer in $\mathcal{C}_F$; equality places it in F_c. If the sets do not intersect, the gap is strict.□

Choose box endpoints by the signs of c and check the group rows. Feasibility proves zero constraint gain. If zero coefficients leave multiple minimizers, one failed point is inconclusive. We apply this test below.

Sampling uncertainty versus ambiguity. Finite views include sampling intervals; population views use exact generating statistics, retaining ambiguity. Among 240 primary queries with true effects above δ, ambiguity blocks 195 and sampling width 33; exact optima exclude optimization failure (Table III). Six additional finite certifications comprise four with sufficient standalone population information and two requiring fusion. Robust design retains the reference in 238/240 lower-budget cases: true effect zero.

For the metadata policy, Fusion adds/loses 6/0, 9/0, 0/2 and 3/1 for $(\Lambda = 2, 100d)$, $(2, 1000d)$, $(\Lambda^*, 100d)$ and $(\Lambda^*, 1000d)$, respectively; oracle Λ^* comes from simulator truth. In the low-budget oracle case, constraint gain averages 0.0114 pp versus 0.0913 pp budget cost. The zero-gain test holds for 193/240 queries, including both losses; the larger-budget loss has positive gain below cost. All three losses persist at exact optima. Different samples and design policies prevent a causal sample-size interpretation; full diagnostics are archived.

a) Standalone-source controls: Observational consistency (5) lets OBS certify 99/100 external 2×2 queries; other policy/view settings differ. KuaiRec's imposed exposures and query-directed randomized means [27] do not establish general complementarity: a 35-setting census finds no true-radius fusion-only case. 400 count/policy pairs remain in the artifact.

A broader 6,000-view coverage audit records 14 misses, all outside the 2,880-view family subset, and no false certification. All 18,900 final-mapping proofs pass checking;

mapping leaves seven group-score certifications unresolved. Design views have one miss in 600. The artifact retains the breakdown.

D. Real data: policy redesign and stopping

Here a policy chooses among 300 coat items for one user population ($K = 1$), without personalization or a ranked list. The Coat dataset provides 24 self-selected and 16 randomly selected item ratings from each of 290 users [1]. A success is a rating of at least four, not a purchase; zero entries denote unobserved ratings. Users, not the 11,600 ratings, are treated as independent. Within each source, potential ratings are assumed unaffected by presentation order or the other items in a user's panel.

Group constraints use *item* attributes: gender category, jacket type and color. Their separate partitions total 31 rows and overlap when combined. Twelve assumption settings cross $\Lambda \in \{1, 1.5, 2, 3\}$ with $\rho \in \{0, .02, .05\}$; the primary setting is $(2, 0)$. Fifteen candidate policies upweight or downweight front-page, men's, fleece, waterproof or black items: $\pi_i \propto t^{h_i}$ for binary attribute h_i and $t \in \{.5, 1.5, 2\}$. The reference selects uniformly $(1/300)$; the primary candidate doubles front-page item weights, then normalizes. Only the three front-page policies have contrasts outside the span of group rows.

To use all ratings, we assume each user's 16 R items form a simple random subset independent of potential ratings. The release describes random/uniform selection; collection code is unavailable. A control retains one rating per source/user without looking at its value, giving $q_a = (16/300)/16 = 1/300$. Its 580 events leave 116 empty O and 124 empty R cells; none of 900 queries or 15 full-cell controls is certified. Discarding the panels loses substantial information.

Inference accounts for repeated ratings. O intervals count included users per item; joint bounds use $N = 290$. Uniform thinning divides inclusion probabilities by 24, preserving their ratio and Λ, but weakens consistency bounds. For R group scores, each user averages their ratings within a group. Under the subset assumption this is an unbiased group-mean estimate; the 72–290 active-user scores lie in $[0, 1]$ and enter (8). Their sample variance uses $n_j - 1$ and requires user-level records. Projected intervals instead use per-item CP and propagate through (9).

Five constructions across twelve settings, plus one full-cell control, give 61 views and $15 \times 61 = 915$ candidate queries. No view is refused or query certified. Under $(\Lambda, \rho) = (2, 0)$, group Fusion improves R group's lower bound by 0.309 pp, but neither passes $\delta = .001$.

In Table IV, Fusion group's -0.01193 bounds one policy's gain, whereas $4/12$ counts assumption settings supporting some qualifying policy, all outside the primary setting. Independent checks reconstruct the statistics, views and proofs.

Stopping under the primary declaration. All five primary views have zero capacity: group Fusion allows every item to have success probability 0.2186 (rounded), making every policy tie with the reference. This certifies that the current view cannot support a guaranteed improvement. The stronger full-cell control also has zero capacity.

TABLE IV
COAT: ONE POLICY VERSUS THE FULL POLICY FAMILY.

	Fixed policy Primary setting	Full policy family Assumption settings		
Construction	**LCB**	$\mathcal{R} = 0$	$0 < \mathcal{R} \le \delta$	$\mathcal{R} > \delta$
OBS	-0.03729	9/12	0/12	3/12
R group	-0.01502	12/12	0/12	0/12
R projected	-0.03402	12/12	0/12	0/12
Fusion group	-0.01193	7/12	1/12	4/12
Fusion projected	-0.03132	9/12	3/12	0/12
R full-cell control	-0.03077	1/1	0/1	0/1

Left: one prespecified policy at $(\Lambda, \rho) = (2, 0)$; lower confidence bounds (LCB) in probability units. Right: counts across twelve assumption settings on the same 290 users; full-cell transport is a stronger, one-setting control. All use $\alpha = .05$ and $\delta = .001$ (0.1 pp).

Searching beyond candidates under other declarations. With ideal uniform reference $1/300$, exact family optimization resolves 60 grid positions: 49 have $\mathcal{R} = 0$, four $0 < \mathcal{R} \le .001$, and seven $\mathcal{R} > .001$. The seven positive positions are five unique requests: OBS at $\Lambda = 1$ repeats across three tolerances; group Fusion passes at $\Lambda = 1$ for all three tolerances and at $\Lambda = 1.5, \rho = 0$. The original fifteen candidates certified none. Each declaration has its own coverage guarantee.

A redesign under stronger assumptions ($\Lambda = 1.5, \rho = 0$). The jacket-type certificate moves all 28 vests' uniform mass to item 0, a men's bomber: its probability rises from $1/300$ to $29/300$; vests receive zero and 271 other items retain $1/300$. All 15 original and 50 extended catalogue policies give every item positive probability. The full family can instead assign zero probability to some items, enabling this redesign. The contrast is $(28/300)(m_0 - \overline{m}_{\text{vest}})$, where $\overline{m}_{\text{vest}}$ is the 28 vests' uniform mean. Its checked lower bound is $(28/300)(\ell_0 - b_{\text{vest}}/\kappa_{\text{vest}})$: row mass κ converts the group constraint to its uniform mean. The only nonzero terms, 2.472638 pp and -2.226283 pp, sum to 0.246355 pp in expected high-rating probability over uniform. This exceeds 0.1 pp and matches the full overlapping view's capacity; matched full-budget OBS and both R views have zero capacity.

Separate gender, jacket-type and color partitions give 144 view memberships: 10 exceed the threshold, 10 have positive capacity at most the threshold, and 124 have zero capacity. Construction and exact LP agree on 95 unique supported requests across the full evaluation; LP also resolves 30 overlaps. Subview capacity cannot exceed its parent's, as relaxation requires.

Under full-cell transport, each user's score is $Z_u = (300/16) \sum_{a \in S_u} c_a Y_{ua}$, with random panel S_u, binary high-rating outcomes Y_{ua} and policy contrast $c = \pi - \pi_0$. The factor $300/16$ corrects each item's inclusion probability $16/300$. A one-sided Hoeffding bound across 290 users uses score range $[\min(0, 300 \min c), \max(0, 300 \max c)]$. The primary

TABLE V
PREPROCESSING THE UNCHANGED 240-QUERY COAT CATALOGUE.

Plan	Full-stream time (s)	Cost relative to Opt	Solver calls N/E/F
Positive-capacity view: $\Lambda = 1.5$			
Hot	6.800	0.974–1.009	2/2/0
Opt	6.934	1.000	2/2/0
Opt+Part	7.312	1.051–1.077	2/2/1
Opt+LP	7.276	1.042–1.079	2/2/1
Zero-capacity view: $\Lambda = 2$			
Hot	6.329	0.985–0.998	1/1/0
Opt	6.390	1.000	1/1/0
Opt+Part	6.431	0.999–1.024	1/1/0
Opt+LP	6.359	0.990–1.002	1/1/0

Each stream has 240 queries. Time: three-round median including startup, preprocessing, verification and cleanup; relative cost: paired range. N/E/F: native/exact-policy/family solver calls per stream, identical across runs. Hot: native hot-start; Opt: optimized threshold execution. Both views use $\rho = 0$.

estimated gain over uniform is 0.00371, with lower bound -0.06452; none of 15 fixed queries passes. These bounds cover each prespecified query separately.

a) Catalogue queries with ordinary caching: A catalogue is a fixed list of candidate policies; each query tests one listed policy at a requested margin. At primary $(\Lambda, \rho) = (2, 0)$, each of five source views receives an extended catalogue combining two indicators: $\pi_i \propto t_1^{h_{1i}} t_2^{h_{2i}}$, $(t_1, t_2) \in \{.5, 1, 2\}^2 \setminus \{(1, 1)\}$. Ten pairs give 80 policy positions at margins $0, .001, .01$ (0, 0.1, 1 pp): 240 requests per view, 50 objectives and 150 objective–margin pairs. Prepared execution retains exact dual terms; optimized-exact adds sparse arithmetic and cached number conversions. Both fully cache objectives and retain a persistent receiver. Native hot-start retains a solver basis without cross-objective certificate candidates. Quality checks precede three timing rounds.

Prepared and optimized-exact agree on all 1,200 requests. Optimized/prepared cost ratios 0.960, 0.941, 0.953 miss the ten-percent target; optimized-exact is slower in 3/15 view-rounds and retains more estimated exact-state payload in every view (not isolated RSS). Native hot-start changes 98 false answers to unknown and four unknown answers to false, never reversing a Boolean answer. These plans therefore do not settle the same set of queries; on R projected, decisions agree and native is faster in all rounds. None certifies improvement, and neither registered fusion-only information target is supported. The constant-action witnesses already establish zero capacity.

b) The cost of family preprocessing: Can family preprocessing reduce catalogue-query cost? We test two 16-row jacket-type Fusion views, $(\Lambda, \rho) = (1.5, 0)$ and $(2, 0)$, selected retrospectively with known capacities. Table V compares Hot, Opt, and Opt with partition (Part) or exact-LP (LP) preprocessing. All share objective caching, checked points and a rational receiver. Family certificates add only feasible points; a shared constant-action check skips unnecessary family solves. Stored-contrast and ideal-policy queries share exact fallback evidence.

All 32 streams complete, each with 240 false answers for both stored and ideal rational policies: none of the listed candidates meets the requested guarantee. At nonzero capacity, Part adds 5.1–7.7% to Opt's full cost without reducing its two native and two exact calls (Table V); LP saves no calls either. At zero capacity, family computation is skipped. The three rounds have incompletely balanced execution positions and different optimal points; costs compare complete preprocessing choices. Family answers enable redesign beyond the catalogue but do not amortize preprocessing here.

VII. RELATED WORK AND SCOPE

Database queries. Causal systems adjust aggregates [28], implement matching [29], define units [30] and explain patterns [31]. Other approaches propagate uncertainty bounds [32] or certify semantic-view predicates from fresh draws [33]. Our source-constrained view supports adaptive objectives under one coverage event and whole-family stopping certificates.

Our view combines binary density-ratio and conditional-moment relaxations [5], [6]. Other fusion methods exploit overlap [34], impose experimental moments [35], or estimate sensitivity frontiers [3]. Propensity-odds policy learning [2], [36] makes different restrictions from our outcome-conditional weights.

Classical allocation. Continuous maximin sharing [37, Sec. 2.2] allocates a common resource across groups. Our groups instead impose disjoint, two-sided constraints on uncertain means; we minimize the best action's advantage over a reference and recover an original-constraint certificate.

Proofs and scope. Filtering [38], assignment caching [39], bounds and reuse are established; VIPR checks exact optimization proofs [13]. IntegriDB authenticates results [40]; our digests bind certificates to a view. Our guarantees cover binary outcomes and known target weights under the stated source assumptions and intervals.

Data ethics. We reanalyze public benchmarks under their release terms, without recruiting or contacting participants. No institutional ethics review was undertaken or additional consent sought.

VIII. CONCLUSION

Q-BRIDGE answers verifiable policy queries over a shared confidence view. Full-family optimization finds a qualifying policy or rules out the requested guarantee. Retaining exact terms lowers group-average cost for repeated queries on the tested streams.

Acknowledgments and AI Usage Declaration

The authors conceived the research, designed the overall study, and led the manuscript writing. OpenAI Codex assisted with drafting, polishing and mathematical exposition (Sections I–VIII), code and experiments (Sections III–VI), figures, and source retrieval. The authors take full responsibility for the manuscript, including any text generated or polished by the LLM, and for ensuring its accuracy, originality, proper attribution, and compliance with publication ethics, including the prevention of plagiarism and scientific misconduct.

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